



Operations Research

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Solution to the Demo Exercises, Sheet 1

Demo Exercises

Exercise D1.1. *Toy production*

Problem

The Toys&Joys GmbH produces wind-up ducks and toy trains made of wood. One wind-up duck can be sold for €27. The material cost per duck is €10, the other variable costs are €14. The selling price for the toy train is €21, the material cost and other variable costs per toy train are €9 and €10, respectively.

Two operations are necessary in the production: Painting and assembling. Painting the individual parts of a duck takes 2 hours, their assembly 1 hour. For the train, painting and assembly each take 1 hour.

Any number of raw materials can be used each week. However, per week a maximum of 100 hours is available for painting, and a maximum of 80 hours is available for assembly. In addition, the demand for wind-up ducks is limited to 40 pieces per week.

Define a linear program whose solution maximizes the weekly profit under the given restrictions. Afterwards, solve this graphically.

Solution

The variables:

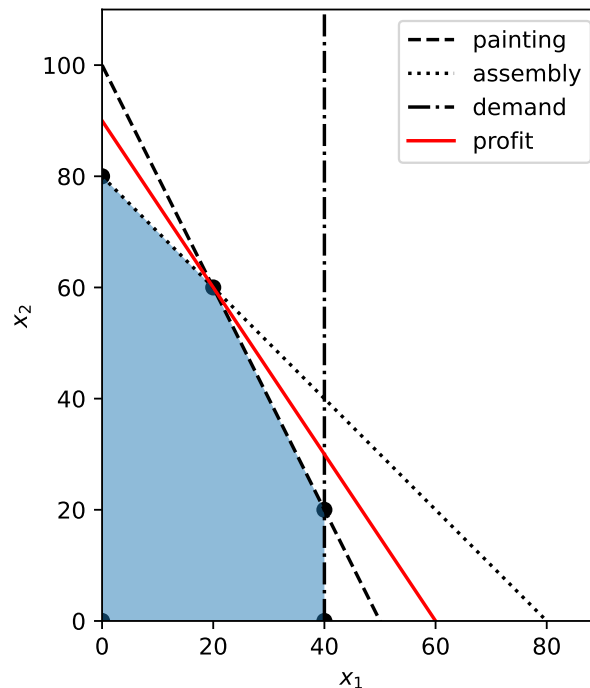
- x_1 : number of wind-up ducks produced per week
 x_2 : number of toy trains produced per week

The model:

Profit = sale revenue – material cost – other variable costs

$$\begin{aligned} \text{Max} \quad & (27 - 10 - 14)x_1 + (21 - 9 - 10)x_2 = 3x_1 + 2x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 100 && \text{capacity restriction painting} \\ & x_1 + x_2 \leq 80 && \text{capacity restriction assembly} \\ & x_1 \leq 40 && \text{demand restriction wind-up ducks} \\ & x_1, x_2 \geq 0 \end{aligned}$$

Graphical solution:



Thus, the toy manufacturer should produce 20 wind-up ducks and 60 toy trains per week to achieve a profit of €180.

Exercise D1.2. Fuel oil trade

Problem

In his spare time, Steffen trades in fuel oil. For this purpose, he owns a tank with q liters capacity, in which he can temporarily store fuel oil for any length of time. For the coming n days, Steffen wants to design a buying and selling plan which maximizes his profit.

On each morning and evening of the n days, a tanker truck with a capacity of r liters arrives at Steffen's house. While Steffen can only buy (but not sell) fuel oil in the morning, he can only sell (but not buy) fuel oil in the evening. Since the tanker truck also targets other customers, it is already filled with g_i liters of fuel oil in the morning of the i -th day and with h_i liters of fuel oil in the evening of the i -th day, before it arrives at Steffen's house.

In addition, the fuel oil price is subject to daily fluctuation. Through extensive research, however, Steffen knows that on day i he can buy fuel oil at a price of b_i euros per liter and sell it for s_i euros per liter.

Steffen can temporarily store at most as much fuel oil as fits into his tank. Moreover, he cannot sell more fuel oil than he actually owns. Besides, a tanker truck can never take more fuel oil than fits into its tank and can sell at most as much fuel oil as it currently has in its tank.

Set up a linear program whose solution is a buying and selling plan which maximizes Steffen's profit. Assume that Steffen's tank is already filled with w liters of fuel oil at the beginning of the first day.

Solution

Variables:

- x_i with $i \in [n]$ liters of fuel oil that Steffen buys on the morning of the i -th day
- y_i with $i \in [n]$ liters of fuel oil that Steffen sells in the evening of the i -th day
- z_i with $i \in [n]$ liters of fuel oil that are in Steffen's tank at the beginning of the i -th day

$$\begin{array}{ll} \text{Max} & \sum_{i=1}^n s_i y_i - \sum_{i=1}^n b_i x_i \\ \text{s.t.} & \end{array} \quad (1)$$

$$x_i \leq g_i \quad \forall i \in [n] \quad (2)$$

$$y_i \leq r - h_i \quad \forall i \in [n] \quad (3)$$

$$z_1 = w \quad (4)$$

$$z_i = z_{i-1} + x_{i-1} - y_{i-1} \quad \forall i \in \{2, 3, \dots, n\} \quad (5)$$

$$z_i + x_i \leq q \quad \forall i \in [n] \quad (6)$$

$$y_i \leq z_i + x_i \quad \forall i \in [n] \quad (7)$$

$$x_i, y_i \geq 0 \quad \forall i \in [n] \quad (8)$$

Explanation:

- (1) Steffen's profit is the difference between his sales revenue and his purchase costs.
- (2) On day i , only as much fuel oil can be purchased as is in the tanker truck.
- (3) On day i only as much fuel oil can be sold as fits into the tank truck.
- (4) At the beginning of the first day, Steffen's tank is filled with w liters of fuel oil.
- (5) At the beginning of the i -th day (with $i \geq 2$), the fill level is equal to that of the last day plus the purchase and minus the sale of the last day.
- (6) On day i , after purchasing new fuel oil, the capacity of Steffen's tank must not be exceeded.
- (7) On day i , only as much fuel oil may be sold as Steffen has in his tank after the purchase.
- (8) The quantities of fuel oil Steffen buys and sells on day i must be non-negative.

Alternative solution:

- x_i with $i \in [n]$ liters of fuel oil that Steffen buys on the morning of the i -th day
- y_i with $i \in [n]$ liters of fuel oil that Steffen sells in the evening of the i -th day

$$\begin{array}{ll} \text{Max} & \sum_{i=1}^n s_i y_i - \sum_{i=1}^n b_i x_i \\ \text{s.t.} & \end{array} \quad (1)$$

$$x_i \leq g_i \quad \forall i \in [n] \quad (2)$$

$$y_i \leq r - h_i \quad \forall i \in [n] \quad (3)$$

$$w + x_1 \leq q \quad (4)$$

$$w + \sum_{i=1}^d x_i - \sum_{i=1}^{d-1} y_i \leq q \quad \forall d \in \{2, 3, \dots, n\} \quad (5)$$

$$w + \sum_{i=1}^d x_i - \sum_{i=1}^d y_i \geq 0 \quad \forall d \in [n] \quad (6)$$

$$x_i, y_i \geq 0 \quad \forall i \in [n] \quad (7)$$

Explanation for the alternative LP:

- (1) Steffen's profit is the difference between his sales revenue and his purchase costs.
- (2) On day i , only as much fuel oil can be purchased as is in the tanker truck.
- (3) On day i only as much fuel oil can be sold as fits into the tank truck.
- (4) On the first day, the capacity of Steffen's tank must not be exceeded.
- (5) On day i , after purchasing new fuel oil, the capacity of Steffen's tank must not be exceeded. The constraint is defined only for days $d \geq 2$, otherwise a y_0 would have to be introduced.
- (6) On day i , only as much fuel oil may be sold as Steffen has in his tank after the purchase.
- (7) The quantities of fuel oil Steffen buys and sells on day i must be non-negative.

Exercise D1.3. *Fitness*

Problem

Georg has signed up at the gym in order to be fully trained by the swimming season. His goal is to train the muscle groups $g = 1, \dots, n$ and to reach a training level of T_g in each of them.

Georg can train up to 1000 minutes until the start of the swimming season. He has the option of either training alone or hiring a personal trainer. He has to pay 1 euro for every minute he is supervised by his personal trainer. The first 400 minutes that Georg trains without a trainer cost him nothing, and for each additional minute that he trains alone, he has to pay 0.10 euros.

In the studio m machines $M_1, \dots, M_m$ are available for Georg to train with. If he trains alone on machine M_j , every minute of training contributes a_{jg} to training level T_g ; every minute of training under the guidance of the personal trainer on M_j contributes p_{jg} .

Create a training plan for Georg so that he achieves his desired level of training in all muscle groups in the time available to him, while minimizing his costs.

Solution

Variables:

- x^a Minutes that Georg trains alone, which exceed his 400 free minutes
- x_j^a Minutes that Georg trains alone on machine M_j
- x_j^p Minutes that Georg trains on machine M_j under the supervision of the personal trainer

$$\begin{array}{ll} \text{Min } 0.1x^a + \sum_{j=1}^m x_j^p & (1) \\ \text{s.t.} & \end{array}$$

$$\sum_{j=1}^m (a_{jg}x_j^a + p_{jg}x_j^p) \geq T_g \quad \forall g = 1, \dots, n \quad (2)$$

$$\sum_{j=1}^m (x_j^a + x_j^p) \leq 1000 \quad (3)$$

$$x^a \geq \sum_{j=1}^m x_j^a - 400 \quad (4)$$

$$x_j^a, x_j^p \geq 0 \quad \forall j = 1, \dots, m \quad (5)$$

$$x^a \geq 0 \quad (6)$$

An alternative formulation of the free minutes goes over the definition of the variables $x_{\leq 400}^a$ and $x_{>400}^a$:

- $x_{\leq 400}^a$ The first 400 minutes that Georg trains alone
- $x_{>400}^a$ The minutes beyond that that Georg trains alone
- x_j^a Minutes that Georg trains alone on machine M_j
- x_j^p Minutes that Georg trains on machine M_j under the supervision of the personal trainer

This changes the LP to

$$\begin{array}{ll} \text{Min } 0.1x_{>400}^a + \sum_{j=1}^m x_j^p & (1) \\ \text{s.t.} & \end{array}$$

$$\sum_{j=1}^m (a_{jg}x_j^a + p_{jg}x_j^p) \geq T_g \quad \forall g = 1, \dots, n \quad (2)$$

$$\sum_{j=1}^m (x_j^a + x_j^p) \leq 1000 \quad (3)$$

$$x_{\leq 400}^a + x_{>400}^a = \sum_{j=1}^m x_j^a \quad (4)$$

$$x_{\leq 400}^a \leq 400 \quad (5)$$

$$x_j^a, x_j^p \geq 0 \quad \forall j = 1, \dots, m \quad (6)$$

$$x_{\leq 400}^a, x_{>400}^a \geq 0 \quad (7)$$

Explanation for the alternative LP:

- (1) Georg pays €0.10 for each minute of training alone, which exceeds his 400 free minutes, and €1 for each minute with his personal trainer.
- (2) For each muscle group g , at least the training level T_g must be reached. Depending on whether Georg trains alone or with a personal trainer on a machine j , the contribution to the training level changes.
- (3) In total Georg must not train more than 1000 minutes.
- (4) The total number of minutes Georg trains alone on the machines must equal the sum of $x_{\leq 400}^a$ and $x_{>400}^a$.
- (5) Georg has a maximum of 400 free minutes.
- (6) The number of minutes Georg trains alone or with a fitness trainer must be non-negative.
- (7) The number of training minutes without a trainer that Georg uses from his free minutes, or with which he exceeds his free minutes, must be non-negative.